THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · EASY LEVEL CHAPTER TEST SET 2

Chapter 11: **Electricity** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Resistance of a metal wire 1

(a) increases with temperature **(b)** is the same at all temperatures

(c) decreases with temperature **(d)** becomes zero on heating

2. In parallel, if one bulb fuses, the others 1

(a) explode **(b)** glow dimmer **(c)** continue to glow **(d)** stop

3. A voltmeter is connected 1

(a) in parallel **(b)** at any place

(c) in series **(d)** in a short circuit

4. The unit of resistivity is 1

(a) ohm **(b)** ohm metre **(c)** metre **(d)** siemens

5. Electric power is the 1

(a) total energy **(b)** resistance

(c) charge **(d)** rate of consumption of energy

6. An ammeter is connected 1

(a) across the cell **(b)** in series **(c)** in parallel **(d)** across the bulb

7. Which is the best conductor? 1

(a) Silver **(b)** Rubber **(c)** Glass **(d)** Nichrome

8. The unit of power is 1

(a) ohm **(b)** watt **(c)** joule **(d)** coulomb

9. The unit of potential difference is 1

(a) ohm **(b)** ampere **(c)** coulomb **(d)** volt

10. The resistivity of an insulator is 1

(a) same as copper **(b)** zero **(c)** very high **(d)** very low

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Define 1 ohm. 2

12. What is the formula for equivalent resistance in parallel? 2

13. What is a potential difference? 2

14. Write the unit of resistivity. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. A potential difference of 12 V is applied across a 4 ohm resistor. Find the current. 3

16. Find the equivalent resistance of 6 ohm and 3 ohm in parallel. 3

17. Why is the resistance of the ammeter kept low? 3

OR

A current of 3 A flows through a conductor having 6 V across it. Find the resistance. 3

18. A 40 W bulb works for 5 h daily. Find the energy used per day in kWh. 3

OR

Why is tungsten used for the filament of an electric bulb? 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Three resistors of 2 ohm, 4 ohm and 6 ohm are connected in series to a 12 V battery. Find the total resistance, current and voltage across each. 5

OR

An electric geyser of 2 kW is used for 3 hours daily. Find the energy used in 30 days and the cost at Rs 5 per kWh. 5

20. Two resistors of 10 ohm and 15 ohm are connected in parallel to a 6 V battery. Find the total resistance and the current in each. 5

OR

A wire of resistance 8 ohm is cut into two equal parts. Find the resistance of each part and the equivalent resistance when they are joined in parallel. 5

END OF QUESTION PAPER Total: 40 Marks

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